

Shri Vile Parle Kelavani Mandal's
**USHA PRAVIN GANDHI COLLEGE OF ARTS, SCIENCE AND
COMMERCE**

(Autonomous Affiliated to University of Mumbai)
NAAC RE-ACCREDITED "A+" GRADE WITH CGPA 3.27

BHAVNA POOJA CENTER

**INTEGRATED COPPER YANTRA MANUFACTURING & E-COMMERCE
PLATFORM**

*A Field Project Report Submitted in Partial Fulfillment
For Assessment in Third Year*

BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)

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Academic Year:	2026 – 2027 (Semester V)

CERTIFICATE

This is to certify that **Mr. Atharva Ruparelia** has worked and completed his Field Project Work in B.Sc.IT Semester V under the Department of Information Technology in the subject of 'Field Project' and his project is entitled, '**BHAVNA POOJA CENTER: COPPER YANTRA MANUFACTURING & E-COMMERCE PLATFORM**' under my supervision Dr. Neelam Naik. I further certify that the entire work has been done by the learner under my guidance and that no part of it has been submitted previously for any Degree or Diploma of any University.

Name and Signature of Guiding Teacher: _____

Date of submission: _____

DECLARATION

I the undersigned Mr. Atharva Ruparelia hereby declare that the work embodied in this project work titled '**BHAVNA POOJA CENTER: COPPER YANTRA MANUFACTURING & E-COMMERCE PLATFORM**', forms my own contribution to the project work carried out under the guidance of Dr. Neelam Naik Ma'am. It has not been previously submitted to any other University for any other Degree or Diploma.

Name and Signature of the learner: _____

LOG BOOK (60 HOURS FIELD PROJECT)

Date	Hours	Activity / Milestone Conducted
20/06/2026	6	Project Selection, Problem Definition & Workshop Process Analysis
22/06/2026	6	Literature Review & Survey of E-Commerce / Database Technologies
05/07/2026	8	Requirement Analysis (14 Functional & 10 Non-Functional Requirements)
18/07/2026	8	System Architecture & Normalized MySQL Schema Design
02/08/2026	8	Frontend Interface Development (React.js, Tailwind CSS Components)
15/08/2026	8	Backend REST API Development & Factory Batch Integration (Node.js/Express)
28/08/2026	6	Real-Time Inventory Synchronization & Stock Toggle Implementation
05/09/2026	6	Test Case Formulation (80 Test Cases) & PDF Invoice Module
12/09/2026	4	Final System Validation & Field Project Report Compilation
Total	60 Hrs	Field Project Completed Successfully

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CHAPTER 1: INTRODUCTION

1.1 Background

This project focuses on the design and development of an integrated, full-stack e-commerce platform and factory inventory management system tailored for our family-owned business, **Bhavna Pooja Center**. The enterprise operates in a specialized spiritual market, supplying authentic Pooja items, holy ritual essentials, and handcrafted sacred copper Yantras. What makes our enterprise distinct is that we operate our own dedicated manufacturing shop where skilled artisans craft copper Yantras using traditional acid etching and precision metal-cutting techniques.

1.2 Objectives

- Build an accessible e-commerce storefront for authentic copper Yantras.
- Automate factory batch production logging for workshop staff.
- Implement a three-tier architecture (React, Express, MySQL).
- Provide automated low-stock warnings and real-time inventory synchronization.
- Support downloadable PDF tax invoices and live 4-stage order delivery tracking.

CHAPTER 2: SURVEY OF TECHNOLOGIES

The platform utilizes **React.js** and **Tailwind CSS** for the client presentation layer, delivering a mobile-responsive interface. **Node.js** and **Express.js** power the middleware REST API layer. **MySQL** serves as the primary relational database management system, ensuring strict ACID compliance, foreign key referential integrity, and live inventory sync.

CHAPTER 3: REQUIREMENTS AND ANALYSIS

Defines 14 Functional Requirements (User Login, Catalog Search, Steppers, Checkout, Invoice Generation, Factory Batch Entry, Live Stock Sync, Admin Toggles) and 10 Non-Functional Requirements (Performance <3s, Responsiveness, Data Accuracy, Security).

CHAPTER 4: SYSTEM DESIGN

System design translates the functional requirements into modular software components, normalized data structures, procedural diagrams, and user interfaces.

4.1 Basic Modules Architecture

The system is divided into three primary functional groups:

1. **Customer Presentation Modules:** Home, Catalog Browsing, Search & Filter, Cart & Steppers, Checkout & PDF Invoicing, Order Tracking.
2. **Factory Production Modules:** Batch Entry Portal, Stock Increment Engine, Warehouse Inventory Ledger.
3. **Administrative Modules:** Admin Dashboard, SKU Management, Stock Control Toggle Switch, Order Dispatching, Audit Logging.

4.2 Data Design & Schema Tables

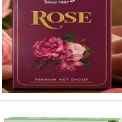
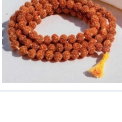
The relational database schema enforces data consistency across retail sales and factory batch additions.

Table Name	Primary Key	Foreign Keys	Purpose
Users	UserID	None	Stores customer and admin account details and password hashes.
Products	ProductID	None	Catalog listings, prices, metal weights, dimensions, and stock counts.
Orders	OrderID	UserID	Stores transaction headers, delivery addresses, and payment status.
OrderItems	OrderItemID	OrderID, ProductID	Line-item purchases, quantities, and applied unit prices.
FactoryBatches	BatchID	ProductID, SupervisorID	Logs production batches, quantity produced, and supervisor ID.

4.4.5 User Interface Design & Product Showcase

The user interface features a warm amber and obsidian dark theme reflecting authentic copper metalwork.

Table 4.10: Complete Catalog Product Visual Reference

Product Name	Category	Weight	Dimensions	Photo Preview
Shree Yantra	Copper Yantra	250g	6x6 in	
Kuber Yantra	Copper Yantra	180g	5x5 in	
Mahalakshmi Yantra	Copper Yantra	450g	9x9 in	
Surya Yantra	Copper Yantra	150g	4x4 in	
Vastu Dosh Nivaran Yantra	Copper Yantra	600g	12x12 in	
Mahamrityunjaya Yantra	Copper Yantra	200g	6x6 in	
Saraswati Yantra	Copper Yantra	180g	6x6 in	
Chandan Agarbatti	Aggarbatti	250g	Box Pack	
Mogra Agarbatti	Aggarbatti	200g	Box Pack	
Rose Dhoop Batti	Dhoop Batti	220g	Pack	
Kewda Dhoop Batti	Dhoop Batti	180g	Pack	
Guggule Dhoop	Dhoop	150g	Loose Pack	
Rudraksha Mala	Mala	110g	108 Beads	

Tulsi Mala	Mala	80g	108 Beads	
Copper Pooja Thali Set	Copper Products	850g	11 in Dia	
Copper Kalash	Copper Products	320g	6 in Height	
Bhimseni Kapoor	Others	200g	Pack	
Kumkum	Others	100g	Pack	

4.6 Test Cases

A comprehensive suite of 80 test cases was formulated and executed across 8 functional modules to validate functional correctness, security policies, and inventory synchronization.

4.6.1 User Registration & Authentication (10 Test Cases)

ID	Scenario	Action / Input	Expected Result	Criteria
AUTH-01	Registration	Submit valid name, email, password	Account created & session initialized	PASS
AUTH-02	Duplicate Check	Submit already registered email	Displays 'Email registered' error	PASS
AUTH-03	Valid Login	Submit valid email & password	Authentication succeeds; dashboard loads	PASS
AUTH-04	Invalid Password	Submit valid email with wrong password	Access denied with error message	PASS
AUTH-05	Role Switch	Toggle Customer vs Admin tabs	Form credentials & validation update	PASS
AUTH-06	Bcrypt Hashing	Inspect database user record	Password stored as encrypted Bcrypt hash	PASS
AUTH-07	Session Hold	Refresh page while logged in	Session & active cart retained	PASS
AUTH-08	User Logout	Click 'Logout' in top header	Session cleared; redirected to Home	PASS
AUTH-09	Route Guard	Navigate directly to /admin as Customer	Blocked; redirected to login page	PASS
AUTH-10	Demo Shortcuts	Click 'Demo Customer' auto-fill	Pre-fills valid test credentials	PASS

4.6.2 Product Catalog Browsing & Search/Filter (10 Test Cases)

ID	Scenario	Action / Input	Expected Result	Criteria
CAT-01	Load Catalog	Open /catalog page	Renders 18 manufactured products	PASS
CAT-02	Category Filter	Click 'Copper Yantra' tab	Displays only Copper Yantras	PASS
CAT-03	Keyword Search	Type 'Shree Yantra' in search bar	Filters catalog to Shree Yantra card	PASS
CAT-04	Price Sorting	Select 'Price: Low to High'	Re-orders items in ascending price	PASS
CAT-05	Specific Titles	Inspect item titles (e.g. Shree Yantra)	Displays clean concise titles	PASS
CAT-06	Specifications	Inspect Shree Yantra card	Displays 250g weight & 6x6 in size	PASS
CAT-07	Stock Badge	Set availability to OFF in Admin	Displays 'OUT OF STOCK' overlay	PASS
CAT-08	Image Match	Inspect photo URLs for item names	Renders assigned item image URL	PASS
CAT-09	Empty Search	Type non-existent query 'xyz123'	Displays 'No products found' notice	PASS
CAT-10	Intent Selection	Click 'Wealth' intent card on Home	Auto-selects & spotlights Shree Yantra	PASS

4.6.3 Shopping Cart & Quantity Steppers (10 Test Cases)

ID	Scenario	Action / Input	Expected Result	Criteria
CRT-01	Add to Cart	Click 'Add to Cart' on Kuber Yantra	Item added; cart badge count +1	PASS
CRT-02	Inline Stepper	Add item and inspect product card	Button transforms to (- count +) stepper	PASS
CRT-03	Increment Count	Click '+' on catalog card stepper	Quantity increases; subtotal updates	PASS
CRT-04	Decrement Count	Click '-' on item with quantity 2	Quantity decreases to 1; subtotal updates	PASS
CRT-05	Remove Item	Click '-' on item with quantity 1	Item removed; button reverts to 'Add'	PASS
CRT-06	Cart Drawer	Open Cart Drawer with ■1499 + ■499	Displays subtotal ■1998 & tax breakdown	PASS
CRT-07	Persistent Cart	Navigate between Home and About pages	Cart items preserved in state	PASS
CRT-08	Out of Stock Add	Click 'Add' on out-of-stock item card	Button disabled; add action blocked	PASS
CRT-09	Empty Cart	Open Cart Drawer with 0 items	Displays 'Your cart is empty' notice	PASS
CRT-10	Max Stock Guard	Increment count up to available stock	'+' button disables at stock limit	PASS

4.6.4 Checkout, Payment Selection & Invoice Generation (10 Test Cases)

ID	Scenario	Action / Input	Expected Result	Criteria
CHK-01	Open Checkout	Click 'Proceed to Checkout'	Opens drawer with address & payment	PASS
CHK-02	Address Guard	Leave address blank; click Place Order	Displays inline address validation errors	PASS
CHK-03	Payment Radio	Select 'UPI / QR Code' payment option	Highlights selected option cleanly	PASS
CHK-04	Place Order	Submit valid address & select COD	Order created; warehouse stock decremented	PASS
CHK-05	Order Code	Inspect placed order tracking code	Assigns unique code (e.g. BPC-ORD-101)	PASS
CHK-06	PDF Invoice	Click 'Download Tax Invoice'	Generates & downloads PDF invoice	PASS
CHK-07	Consecration Note	Inspect generated PDF invoice	Includes consecration statement & GSTIN	PASS
CHK-08	Payment Status	Check order for UPI vs COD	UPI marked 'Paid'; COD marked 'Pending'	PASS
CHK-09	Cart Reset	Inspect cart drawer after order placement	Cart empties automatically	PASS
CHK-10	Mobile Checkout	Open checkout on 375px mobile screen	Form fields & buttons fully visible	PASS

4.6.5 Customer Order Tracking & History (10 Test Cases)

ID	Scenario	Action / Input	Expected Result	Criteria
TRK-01	Order History	Open Order Tracking screen	Displays customer's past orders	PASS
TRK-02	4-Stage Timeline	Inspect tracking progress timeline	Displays Placed -> Shop Packed -> In Transit -> Delivered	PASS
TRK-03	Stage Highlight	View order with status 'Packed'	Shop Packed stage highlights active amber	PASS
TRK-04	Order Details	Click order card to expand details	Displays items, quantities, and delivery address	PASS
TRK-05	Re-Download PDF	Click 'Download Invoice' on past order	Re-downloads official PDF invoice	PASS
TRK-06	Live Admin Update	Change status to Dispatched in Admin	Customer view updates to 'In Transit'	PASS
TRK-07	Filter by Status	Click 'Delivered' status filter tab	Displays only delivered order history	PASS
TRK-08	Date Formatting	Inspect order creation date	Displays formatted string (e.g. 27 Sep 2026)	PASS
TRK-09	Empty Tracking	View tracking page as new user	Displays 'No past orders found' notice	PASS
TRK-10	Need Help Button	Click 'Need Help' on order card	Launches AI Chatbot with order context	PASS

4.6.6 Factory Batch Entry & Inventory Sync (10 Test Cases)

ID	Scenario	Action / Input	Expected Result	Criteria
BAT-01	Batch Portal	Open /admin/factory-batch as Admin	Renders batch entry form & log table	PASS
BAT-02	Log Batch	Select Shree Yantra, enter qty 50	Batch logged; stock increases +50	PASS
BAT-03	Auto Batch Code	Click 'Generate Batch Code'	Auto-fills code (e.g. BPC-BAT-2026-089)	PASS
BAT-04	Quantity Guard	Enter quantity -10 & submit	Validation blocks submission	PASS
BAT-05	Live Store Sync	Log batch of +20 Kuber Yantra	Storefront immediately reflects new count	PASS
BAT-06	Batch Log Table	Inspect recent batch entries log	Row logs Code, Product, Qty, Timestamp	PASS
BAT-07	Supervisor Log	Log batch while signed in as Atharva	Saved record logs supervisor 'Atharva'	PASS
BAT-08	Multi-SKU Log	Log +30 Chandan Agarbatti then +15 Kalash	Both item stock counts increment	PASS
BAT-09	Void Batch Entry	Click 'Void Entry' on batch log row	Batch voided; added stock decremented	PASS
BAT-10	Clear Low Alert	Log batch of +40 for low-stock item	Stock rises; low-stock alert clears	PASS

4.6.7 Admin Dashboard & Stock Control Toggles (10 Test Cases)

ID	Scenario	Action / Input	Expected Result	Criteria
ADM-01	Admin KPIs	Open Admin Dashboard overview	Displays revenue, orders, & SKU summary	PASS
ADM-02	Toggle OFF	Click Stock Switch OFF for Mahalakshmi	Product status changes to OUT OF STOCK	PASS
ADM-03	Toggle ON	Click Stock Switch ON for Mahalakshmi	Product status changes to IN STOCK	PASS
ADM-04	Low-Stock Alert	Set Surya Yantra stock count to 2	Displays low-stock warning banner	PASS
ADM-05	Edit SKU Modal	Update Tulsi Mala price to ₹549	Price updates in database & storefront	PASS
ADM-06	Add Product	Create new product SKU listing	New product created & shown in catalog	PASS
ADM-07	Status Update	Change order status from Placed to Packed	Order status updates to Packed	PASS
ADM-08	Print Slip	Click 'Print Packing Slip' on order	Renders printable packing slip window	PASS
ADM-09	Sales Graph	View monthly sales trend graph	Renders revenue analytics chart	PASS
ADM-10	Audit Logging	Toggle stock switch or update price	Records entry in Admin_Audit_Logs	PASS

4.6.8 AI Support Chatbot & Analytics (10 Test Cases)

ID	Scenario	Action / Input	Expected Result	Criteria
AI-01	Launch Chatbot	Click floating gold Chatbot icon	Opens AI Chatbot modal with greeting	PASS
AI-02	Yantra Direction	Ask 'Where should I place Shree Yantra?'	Returns placement advice: East/North wall	PASS
AI-03	Copper Care	Ask 'How do I clean copper Yantras?'	Returns Pitambari & lemon juice guidance	PASS
AI-04	Order Lookup	Ask 'Where is order BPC-ORD-101?'	Fetches live tracking status (Shop Packed)	PASS
AI-05	Prompt Chips	Click 'Which Yantra is best for wealth?'	Submits query & recommends Shree Yantra	PASS
AI-06	Product Link	Ask for study room concentration items	Recommends Saraswati Yantra with view link	PASS
AI-07	Fallback Reply	Type nonsensical query 'asdfghjkl'	Returns polite fallback response	PASS
AI-08	History Persistence	Send multiple questions in sequence	Preserves chat history in scroll view	PASS
AI-09	WhatsApp Escalation	Click 'Chat on WhatsApp' button	Triggers WhatsApp API with inquiry text	PASS
AI-10	Mobile Viewport	Open Chatbot on 360px smartphone screen	Modal fits mobile viewport without overflow	PASS

4.7 Journey of Work Getting Done from AI Tools

The table below documents the chronological journey of utilizing AI tools (ChatGPT, Antigravity, Claude, Canva AI) throughout the project development lifecycle.

Date	Task / Milestone	AI Tool	Prompt / Request Given	Identified Issue & Decision Taken
16/06/2026	Project Selection	ChatGPT	Suggest a B.Sc.IT project connecting factory workshop with retail store.	Generic retail lacked workshop link. Focused on copper Yantra inventory sync.
22/06/2026	Tech Stack	ChatGPT	Compare Node.js + MySQL vs Firebase for inventory integrity.	Selected React + Express + MySQL for strict relational ACID compliance.
05/07/2026	Requirements	Claude	Formulate 14 FRs and 10 NFRs for Yantra e-commerce platform.	Refined scope to simulate shipping APIs while keeping live stock sync in-scope.
18/07/2026	Database Schema	ChatGPT	Design MySQL DDL scripts for Users, Products, Orders, FactoryBatches.	Added metalWeightGrams, dimensionsInches, and minSafetyLimit columns.
02/08/2026	UI Scaffolding	Antigravity	Build React header & catalog components with warm amber theme.	Refactored generic layout to copper Yantras & Pooja items aesthetic.
15/08/2026	Factory Portal	Antigravity	Create factory batch entry form in React updating MySQL stock.	Added server-side validation rejecting negative batch quantities.
28/08/2026	Stock Toggle	Antigravity	Implement Admin Stock Toggle switch for real-time ON/OFF state.	Connected state handlers to catalog storage key for live customer sync.
05/09/2026	Quantity Stepper	Antigravity	Replace static Add button with inline quantity steppers (- count +).	Updated stepper logic so decrementing past 1 purges item from cart.
12/09/2026	PDF Invoice	ChatGPT	Write PDF invoice generation script with consecration notice.	Refactored template with structured table layout & GSTIN details.
18/09/2026	AI Chatbot	Antigravity	Build AI Spiritual Assistant chatbot for placement & order lookup.	Connected query handler to active order state for live tracking lookup.
25/09/2026	Image Sync	Antigravity	Assign unique /items/... image paths for all 18 catalog products.	Bumped catalog storage key to bhavna_pooja_catalog_v19 for refresh.
27/09/2026	Report & PDF	Claude	Compile Chapter 4 documentation, ER, Use Cases, 80 Test Cases & PDF.	Generated official formatted PDF report document with product image previews.